Tides Through the RG Window

Exact Forcing, Real-Gauge Evidence, and What Harmonic Prediction Cannot See

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Abstract

Tides are a clean test for RG. A hidden source, the Moon and Sun, produces an exactly computable differential field. The sea transforms that field through a basin that the record never fully reveals. We lock the forcing (Newtonian, exact) and prove five identification theorems. We then test RG on one year of real hourly sea level: the UTide example station `can1998`, 8,760 hours in 1998.

Forcing.

- The differential acceleration is 1.12828, 1.07355 and $-0.54993\ \mu\mathrm{m\,s^{-2}}$ at the sub-lunar point, the far side and quadrature.
- The quadrupole has eigenvalues $\frac{\mu}{D^3}(2,-1,-1)$. The equilibrium heights are $+0.357$ and $-0.179\,\mathrm{m}$.
- The Sun's quadrupole strength is 0.459 times the Moon's, and the idealized M2 period is 12.4206 h.
- The quadrupole sees the lunar *axis*, not its direction: $\mathbb{RP}^2$. Odd-degree terms break this, and the station's M3 constituent ($6.8 \pm 0.3\,\mathrm{cm}$) is such a parity-breaking channel.

The theorems.

(i) *Uniqueness:* a damped linear shallow-water response is unique for fixed forcing, basin and initial data (energy method).

(ii) *Basin fibre:* tidal predictions depend on the basin only through its admittance at the resolved constituent frequencies. Two stable rooms that agree there to machine zero differ by up to 37 off-band, for instance at 3 cpd.

(iii) *Rayleigh resolution:* the K1/P1 Gram condition is 6.9 for 29 days and 1.0 for a year, and UTide drops P1 at 29 days. "Inference" fills that fibre by a declaration. At this station the fitted P1/K1 is 0.403 against an equilibrium 0.331 (22% off), and K2/S2 is 0.318 against 0.272.

(iv) *Prediction ceiling:* no predictor built from tidal constituents beats the non-tidal floor. Out of sample (fit January–June, predict July–December), skill is 0.762 and RMSE 0.217 m against a floor of 0.172 m: 80% of the error is a component the tidal protocol cannot see. 88% of the residual variance is subtidal.

(v) *Hidden, not null:* the residual has lag-1 autocorrelation 0.957. The record's own memory cuts the 1-hour error from 0.217 to 0.062 m and the 6-hour error to 0.143 m, returning to tide-only by 4–7 days. A naive i.i.d. 95% band covers 91.9%.

What RG adds. Each element is known to tidal science. What RG adds is one ledger: which numbers are *determined*, which are *declared* (inference, nodal factors, equilibrium phases), and which are *hidden* (the non-tidal sea, the off-band basin), with a sharp record minimum and a prediction ceiling. It supplies no extra force.

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1 The puzzles

- Why are there two bulges, including one on the side away from the Moon?
- Why is high water not under the Moon?
- Why do tide tables fail in storms?
- Why do short records need "inference"?
- Why can two different harbours have the same tidal constants?
- Can a sealed room detect the Moon?

Each puzzle is a question about *what a record determines*. RG's answer separates the source (exact), the basin (the hidden room), and the record (protocol-relative).

2 The forcing, exactly

Theorem 2.1 (Differential forcing and its quadrupole)**.** *In non-rotating axes translating with Earth's centre, with* $\mathbf{R}$ *the Moon's position and* $\mu_M = GM_M$*, the differential acceleration is*

$$\mathbf{a} = \mu_M\Big[\frac{\mathbf{R}-\mathbf{r}}{|\mathbf{R}-\mathbf{r}|^3} - \frac{\mathbf{R}}{|\mathbf{R}|^3}\Big] = T\mathbf{r} + O\Big(\frac{\mu_M|\mathbf{r}|^2}{D^4}\Big), \qquad T = \frac{\mu_M}{D^3}(3\mathbf{n}\mathbf{n}^\top - I),$$

with eigenvalues $\frac{\mu_M}{D^3}(2,-1,-1)$*. The equilibrium height of a static global ocean on a rigid Earth is*

$$\eta_{\mathrm{eq}} = \frac{\mu_M R_E^2}{gD^3} P_2(\cos z).$$

Proof. Expand $|\mathbf{R}-\mathbf{r}|^{-1}$ in Legendre polynomials. The constant and linear terms are removed by the subtraction of Earth's own acceleration. The gradient of the degree-2 term gives T. □

Computed (F1), with nominal constants.

- Sub-lunar point: 1.12828×10^{-6}; far side: 1.07355×10^{-6}; quadrature: $-5.4993 \times 10^{-7}\,\mathrm{m\,s^{-2}}$.
- Equilibrium heights: $+0.3573$ and $-0.1786\,\mathrm{m}$.

- Sun/Moon ratio: 0.4590.
- Idealized M2 period: 12.4206 h.

The far-side bulge needs no repulsion: the Moon pulls Earth's centre harder than the far-side water.

Theorem 2.2 (The quadrupole sees an axis, not a direction)**.** *$T(\mathbf{n}) = T(-\mathbf{n})$, so a perfect quadrupole record identifies the lunar direction only in $S^2/\{\pm\} = \mathbb{RP}^2$. With known μ_M, the top eigenvalue gives $D = (2\mu_M/\lambda_{\max})^{1/3}$; with unknown μ_M, only μ_M/D^3 is identified. The odd-degree (P_3) potential breaks the antipodal symmetry.*

At this station the third-diurnal M3 constituent is 0.068 ± 0.003 m, far above its equilibrium size (basin amplification). It is the record's parity-breaking channel, which makes this hidden ambiguity *resolvable*, not null.

Remark 2.3 (A sealed room). Light can be blocked, but a tidal gradient cannot. Separated free-falling masses inside the room measure T. A uniform field is undetectable from inside, a tidal one is not. This is the operational content of "curvature" (the electric part of the tidal tensor).

3 The sea as a hidden room

Theorem 3.1 (Conditional uniqueness)**.** *For the linear rotating shallow-water model*

$$\partial_t \mathbf{u} + f\mathbf{k} \times \mathbf{u} = -g\nabla\eta + \nabla U - r\mathbf{u}, \qquad \partial_t \eta + \nabla \cdot (H_b \mathbf{u}) = 0,$$

with no-normal-flow boundaries and $r \geq 0$, two regular solutions with the same forcing, basin and initial data coincide.

Proof. The difference energy $E = \frac{1}{2}\int (H_b|\Delta\mathbf{u}|^2 + g|\Delta\eta|^2)$ satisfies $\dot{E} = -\int r H_b |\Delta\mathbf{u}|^2 \leq 0$, because the Coriolis term does no work. Since $E(0) = 0$, $E \equiv 0$. □

The response is *determined* by forcing and basin. The record, however, does not determine the basin.

Theorem 3.2 (Basin fibre of harmonic constants)**.** *Let a stationary record contain the constituent frequencies $\omega_1, \ldots, \omega_K$. Every tidal prediction made from it depends on the basin only through the complex admittances $Z_k = H(\omega_k)$. Two linear rooms with $H_A(\omega_k) = H_B(\omega_k)$ for all k produce identical tidal records of any length, and hence identical tide tables. Off the constituent frequencies they may differ arbitrarily.*

Proof. A stationary tidal forcing is a sum of sinusoids at the ω_k. A linear response maps each one through $H(\omega_k)$.

Construction (D7).

- Room A has ten damped tidal-band modes, whose real couplings are solved exactly from the five fitted admittances (O1, K1, N2, M2, S2).
- Room B is A plus a stable filter with real impulse response, $Q(\omega) \propto \prod_k (\omega_k^2 - \omega^2) / \prod_p (\omega_p^2 - \omega^2 + 2i\gamma\omega)$, which vanishes at every constituent.

The mismatch at the constituents is 0 (exactly). Off band, at 0.2–4 cpd, the responses differ by up to 37, for example $-4.47 - 0.42i$ against $4.82 - 36.3i$ at 3 cpd. □

Corollary 3.3 (What tide tables are blind to)**.** *Harmonic constants are not a basin model. A changed forcing at other frequencies (storm surge, seiches), or a changed basin (dredging, sea-level rise altering depth and resonance), lies outside what they determine. Predicting it needs an added protocol: a hydrodynamic model with bathymetry, or a meteorological forcing channel.*

4 Resolution, inference and declared fibres

Proposition 4.1 (Rayleigh resolution as identifiability)**.** *Two sinusoids with frequency difference $\Delta\omega$, observed for a time T, have a Gram matrix whose conditioning grows without bound as $\Delta\omega T \to 0$. For K1 and P1 the Rayleigh period is* 182.6 *days. The exact design conditioning is* 6.92 *for 29 days,* 1.0012 *for 182.6 days and* 1.0007 *for 365 days (D5). UTide's automatic selection drops P1 for 29 days and keeps it for a year.*

Proposition 4.2 (Inference is a declaration filling a fibre)**.** *In a short record, K1 and P1 form one fibre. Standard practice fills it by inference: P1 is set from K1 by the equilibrium ratio and phase, i.e. the "credo of smoothness" of the admittance [1]. RG types this as* declared*. At this station the full-year fit gives:*

- *P1/K1* $= 0.403$ *against the equilibrium* 0.331*, with phase difference* $4.8°$*;*
- *K2/S2* $= 0.318$ *against* 0.272*.*

Replacing the fitted P1 by the inferred one changes the full-year prediction by 3.6 *mm rms (D6). The cost is small, but it is a declaration, not an observation.*

Also declared. Nodal corrections (the 18.6-year modulation, unresolvable in one year) and equilibrium phase arguments are supplied by astronomy, not by the record.

5 Prediction power: the ceiling and the memory

Theorem 5.1 (Prediction ceiling for tidal-only predictors)**.** *For any predictor built from the declared constituent set, the out-of-sample mean-square error is at least the residual of the best constituent fit on the test interval itself.*

Proof. Least squares on the test interval is the minimum over that predictor class. □

Real data (D1, D2), station can1998.

- Full-year harmonic analysis: 59 constituents explain 79.1% of the variance. M2 = 0.371, S2 = 0.243, O1 = 0.109, K2 = 0.077 and M3 = 0.068 m, each with 95% half-width about 0.003 m.
- Out of sample, fitting January–June and predicting July–December: skill 0.762, RMSE 0.217 m.
- The oracle floor for the test half is 0.172 m. Better tidal analysis can remove at most 0.044 m. About 80% of the prediction error belongs to a component that the tidal protocol cannot see.

The hidden component (D4). The residual rms is 0.180 m. Of its variance, 87.7% is subtidal (< 0.5 cpd, i.e. weather and shelf dynamics), 1.4% diurnal and 2.2% semidiurnal.

Theorem 5.2 (Hidden, not null)**.** *The non-tidal component is invisible to the tidal protocol, but it is a hidden value of the record's history. Its lag-1 autocorrelation is* 0.957*. Adding the record's memory, an AR(1) nowcast on the residual, recovers it up to its decorrelation time.*

Horizon (h)	1	3	6	12	24	48	96	168
Tide only (m)	0.217	0.217	0.217	0.216	0.216	0.217	0.217	0.217
Tide + memory (m)	**0.062**	0.121	0.143	0.147	0.182	0.211	0.216	0.217

Calibration (D3). A $\pm 1.96\sigma$ band built from the training residual covers 91.9% of the test hours, not 95%. The residual is strongly dependent, and an i.i.d. band is blind to that dependence. RG requires the dependence to be declared before an interval is issued.

6 What current tide models lack, and what RG supplies

Standard practice	Blind spot	RG object
Harmonic constants as the harbour's tide	not a basin model (Theorem 3.2)	basin fibre; added protocol needed for changed forcing or basin
Inference for unresolved pairs	declared, not observed (P1/K1 off by 22% here)	declaration ledger; Rayleigh-sharp minimum record
Nodal corrections, equilibrium phases	astronomy fills an unresolvable fibre	typed as declared
Tide table as the forecast	non-tidal floor 80% of the error	prediction ceiling; hidden component
i.i.d. residual bands	91.9% coverage, autocorrelation 0.957	dependence declared before intervals
Single-frequency response fits	exact rescaling fibre; one frequency cannot separate rooms	multi-frequency protocol separates
Quadrupole-only interpretation	axis, not direction ($\mathbb{RP}^2$)	odd-degree channel (M3) resolves

Honest scope. Every row is known somewhere in tidal science: the response method and credo of smoothness, the Rayleigh criterion, inference, nodal factors, surge modelling and residual analysis [1, 2, 3]. RG's contribution is one typed ledger, determined, declared and hidden, together with certificates. These are a sharp record minimum, a ceiling, and the memory test. RG supplies no additional force.

7 Ledger

Claim	Status	Basis
Differential forcing, quadrupole, equilibrium heights	DERIVED (Newtonian)	Theorem 2.1, F1
Axis but not direction ($\mathbb{RP}^2$); odd terms resolve	EXACT	Section 2; M3 in data
Linear damped response is unique	CONDITIONAL PROOF	energy method
Harmonic constants determine only $\{H(\omega_k)\}$	THEOREM + construction	Theorem 3.2, D7
K1/P1 unresolvable in 29 days; inference is declared	EXACT + REAL DATA	D5, D6
Prediction ceiling; 80% of error non-tidal	THEOREM + REAL DATA	D2
Non-tidal is hidden, not null; memory recovers it	REAL DATA	D8
Coastal sea level at a named place and date	NOT COMPUTED	needs ephemerides, bathymetry, meteorology

8 Data, reproduction and limits

- **Data.** `can1998.dtf`, hourly sea level for 1998, distributed as the real-data example of UTide (`github.com/wesleybowman/UTide`, notebooks). It is analysed at latitude $-25°$ as in that example. The SHA-256 is recorded. Flags of 2 are set to missing and interpolated, as in the example.
- **Code.** `rg_tide_real.py` (NumPy, pandas, UTide 0.4.0) runs F1 and D1–D8. Reruns are byte-identical.
- **Limits.**
  - One station and one year. The station's identity and datum are as distributed.
  - Admittance phases are relative to the Greenwich equilibrium argument (longitude not used), which does not affect the fibre construction.
  - No meteorological data, so the non-tidal component is typed, not explained.
  - The D7 rooms are mathematical constructions consistent with the record, not models of this basin.
  - The AR(1) nowcast is the simplest memory channel, not an operational surge forecast.

References

[1] W. H. Munk, D. E. Cartwright, Tidal spectroscopy and prediction, *Phil. Trans. R. Soc. A* 259 (1966) 533.

[2] R. Pawlowicz, B. Beardsley, S. Lentz, Classical tidal harmonic analysis including error estimates in MATLAB using T_TIDE, *Comput. Geosci.* 28 (2002) 929.

[3] M. G. G. Foreman, *Manual for Tidal Heights Analysis and Prediction*, Institute of Ocean Sciences, Pacific Marine Science Report 77-10, 1977 (revised 2004).

[4] D. L. Codiga, *Unified Tidal Analysis and Prediction Using the UTide Matlab Functions*, GSO Technical Report 2011-01, University of Rhode Island, 2011.

[5] D. E. Cartwright, R. J. Tayler, New computations of the tide-generating potential, *Geophys. J. R. Astron. Soc.* 23 (1971) 45.

[6] M. Rada, *Resolution Geometry* (print edition) and companions: Wall-Tap Structural; RG Topology; Perfect Window, 2026.